

MATH102 SMP Week 6

2020F Mid Problem 7

Find all the critical points of

$$f(x, y) = 3y^2 - 2y^3 - 3x^2 + 6xy$$

And, for each critical point, determine whether it is a local maximum point, a local minimum point, or a saddle point.

2020F Mid Problem 8

A wire 120cm long is cut to three pieces of length x , y and $120 - x - y$, and each piece is bent into the shape of a square. Let $f(x, y)$ denote the sum of the areas of these squares. Find the absolute minimum of $f(x, y)$.

2020F Mid Problem 9

Use the Taylor series of $f(x, y) = e^x \cos y$ to find the quadratic approximation at the origin $(0, 0)$.

2020F Mid Problem 10

Find the maximum value and the minimum value of xyz in $x^2 + y^2 + z^2 \leq 9$.

2020F Mid Problem 11

For each $N \in \mathbb{N}$, define the set $\mathcal{A}_N$ by

$$\mathcal{A}_N := \left\{ \sum_{k=0}^N r_k \vec{a}_k : r_k \geq 0 \forall k = 0, \dots, N, \sum_{k=0}^N r_k \leq 1 \right\}$$

where $\vec{a}_k = (\cos(\frac{\pi}{6} + \frac{2k\pi}{N}), \sin(\frac{\pi}{6} + \frac{2k\pi}{N}))$ for each $k = 0, \dots, N$. Let α_N and β_N be the maximum value and the minimum value of $f(x, y) = x + 5y$ on the region $\mathcal{A}_N$, respectively. Find the limit

$$\lim_{N \rightarrow \infty} \alpha_N \beta_N$$